

Nanyang Technological University
SPMS/Division of Mathematical Sciences

AY20/21 MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS Test 1

Name &
Matriculation Number

Score:

Venue:

HINTS

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1. (40 Points) A factory finds that, on average, 20% of the bolts produced by a given machine will be defective for certain specified requirements. If 10 bolts are selected at random from the day's production of this machine, find the probability

- (a) that exactly 2 will be defective;
- (b) that 2 or more will be defective.

Hints:

- (a) Let A denote the event that exactly 2 among the 10 selected bolts are defective. Then

$$P(A) = \binom{10}{2} (0.20)^2 (1 - 0.20)^8 \approx 0.302.$$

- (b) Let B denote the event that 2 or more among the 10 selected bolts are defective. Then

$$P(B) = 1 - P(B') = 1 - \binom{10}{0} (0.20)^0 (1 - 0.20)^{10} - \binom{10}{1} (0.20)^1 (1 - 0.20)^9 \approx 1 - 0.107 - 0.268 = 0.624.$$

2. (30 Points) A laboratory blood test is 95% effective in detecting a ceratin disease when it is, in fact, present. However, the test also yields a “false positive” result for 1% of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply that he or she has the disease.) If 0.5% of the population actually has the disease, what is the probability a person has the disease given that the test result in positive?

Hints: Let A denote the event that the test shows positive, and B the event that a person has the disease. Then B' is the event that a person does not have the disease. We have

$$P(A|B) = 0.95, \quad P(A|B') = 0.01, \quad P(B) = 0.005, \quad P(B') = 0.995.$$

Then

$$\begin{aligned} P(B|A) &= \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')} \\ &= \frac{(0.95)(0.005)}{(0.95)(0.005) + (0.01)(0.995)} \\ &\approx 0.3231. \end{aligned}$$

This is the probability a person has the disease given that the test result in positive.

3. (30 Points) The number of eggs laid on a tree leaf by an insect of a certain type is a Poisson random variable with parameter λ . However, such a random variable can be observed only if it is positive, since if it is 0, then we cannot know that such an insect was on the leaf. If we let Y denote the observed number of eggs, then

$$P\{Y = i\} = P\{X = i | X > 0\}$$

where X is Poisson with parameter λ . Find $\mathbb{E}(Y)$.

Hints:

$$\begin{aligned} P\{Y = i\} &= \frac{P\{X = i \mid X > 0\}}{P\{X = i \text{ and } X > 0\}} \\ &= \frac{P\{X = i\}}{P\{X > 0\}} = \frac{\frac{\lambda^i e^{-\lambda}}{i!}}{1 - e^{-\lambda}} = \frac{\lambda^i e^{-\lambda}}{i!(1 - e^{-\lambda})}. \end{aligned}$$

Thus,

$$\begin{aligned} \mathbb{E}(Y) &= \sum_{k=1}^{\infty} k \times \frac{\lambda^k e^{-\lambda}}{k!(1 - e^{-\lambda})} \\ &= \frac{1}{1 - e^{-\lambda}} \sum_{k=1}^{\infty} \frac{k \lambda^k e^{-\lambda}}{k!} = \frac{1}{1 - e^{-\lambda}} \mathbb{E}(X) = \frac{\lambda}{1 - e^{-\lambda}}. \end{aligned}$$